INFOSYS SE — MATCHSTICK, CUBE & SPATIAL PUZZLES

6 Rigorous Spatial Questions • Minimal Matchstick Removals, 7×7×7 Cube Paint Cuts & Triangle Counting

SECTION 1: Matchstick, Cube & Spatial Logic Puzzles (6 Questions)

[College ITM Mock Paper 2]

Directions: *Matchstick Grid Puzzle: A row of 5 unit squares is built using 16 matchsticks.*

Q1. What is the minimum number of matchsticks to remove so that exactly 3 complete unit squares remain?

a) 1
b) 2
c) 3
d) 4

Answer: A

Logic / Trick: The matchstick shared between the 2nd and 3rd unit squares is a side of both. Removing that single internal stick breaks both squares at once, leaving the 1st, 4th, and 5th squares intact (exactly 3 complete squares) with only 1 stick removed.

[College ITM Mock Paper 3]

Directions: *Matchstick L-Shape Puzzle: A figure is made of 3 unit squares arranged in an L-shape using 10 matchsticks in total.*

Q2. What is the minimum number of sticks to remove so that no complete square remains?

a) 1
b) 2
c) 3
d) 4

Answer: B

Logic / Trick: The stick shared between the top-left square and top-right square is a side of both. Removing it breaks both squares in one move. Removing one more exclusive stick of the bottom square breaks the third. Minimum sticks = 2.

[College ITM Mock Paper 5]

Directions: *Matchstick Plus-Sign Puzzle: A cross figure is made of 5 unit squares (1 central square + 4 outer squares attached) using 16 matchsticks.*

Q3. What is the minimum number of sticks that must be removed so that no complete square remains?

a) 3
b) 4
c) 5
d) 6

Answer: B

Logic / Trick: Each of the 4 outer squares shares exactly one stick with the centre square and no sticks with each other. Removing all 4 shared inner sticks breaks all 4 outer squares and simultaneously destroys the centre square as well. Minimum = 4.

[College ITM Mock Paper 6]

Directions: *Matchstick 2x3 Grid: A 2 x 3 grid of unit squares is built from 17 matchsticks and contains 8 squares in total (six 1x1 squares and two 2x2 squares).*

Q4. What is the minimum number of sticks that must be removed so that no square of any size remains?

a) 3
b) 4
c) 5
d) 6

Answer: B

Logic / Trick: Exhaustive verification over all 17 sticks and 8 squares confirms that removing 3 sticks leaves at least one 1x1 or 2x2 square intact. Removing a specific set of 4 sticks destroys all six 1x1 and both 2x2 squares. Minimum = 4.

[College ITM Mock Paper 6]

Directions: *Cube Painting & Cutting: A solid wooden cube of side 7 units is painted on all 6 faces and then cut into 1-unit smaller cubes (7 x 7 x 7 = 343 small cubes).*

Q5. How many small cubes have at most 1 painted face?

a) 275
b) 125
c) 150
d) 335

Answer: A

Logic / Trick: Total cubes = 7^3 = 343. "At most 1 painted face" = (0 painted faces) + (1 painted face). Cubes with 0 painted faces = (7-2)^3 = 5^3 = 125. Cubes with 1 painted face = 6 * (7-2)^2 = 6 * 25 = 150. Sum = 125 + 150 = 275.

[College ITM Mock Paper 1]

Directions: *Geometry & Figure Counting: In triangle ABC, medians AD, BE and CF are drawn and meet at centroid G.*

Q6. How many triangles are there in the figure in total?

a) 12
b) 14
c) 16
d) 18

Answer: C

Logic / Trick: 6 small triangles meeting at centroid G. 3 triangles formed by pairing adjacent small triangles across medians (taking 2 at a time = 6). 3 large triangles divided by each median (taking 3 at a time = 3). Plus the whole triangle ABC = 1. Total = 6 + 6 + 3 + 1 = 16.